

SIMULTANEOUS *Go* VIA QUANTUM COLLAPSE

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ABSTRACT. We construct a simultaneous combinatorial game *SGo* that is in a precise mathematical sense a symmetric simultaneization of the classical board game *Go*. We develop some general theory of this symmetric simultaneization here, which may be of independent interest. *SGo* is in some ways a simpler game than *Go*, as Komi, Ko and suicide rules are removed. On the other hand it has similar dynamics, enabled by certain deterministic “quantum state” reduction, so that state evolution is deterministic. Using the argument of Nash, we show that a symmetric simultaneous combinatorial game has a mixed equilibrium drawing strategy, and so *SGo* does as well.

1. INTRODUCTION

Go is an ancient game governed by local-to-global geometric principles. It is a sequential game (that is, turns alternate between the two players), which has the benefit of a very simple theoretical structure, (mostly) fitting into the subject of combinatorial game theory. However, this structure also has drawbacks: we need an ad hoc rule such as Komi to deal with the advantage of Black having the first turn. Moreover, the Ko situation and its cousins like triple Ko necessitate certain non repetition rules. Although non repetition rules are necessary in the end game of many types of games, like chess for example, in *Go* they are necessary even in the early game, which is to say the least inelegant. Furthermore, this leads to Ko fights, in which the game loses its local-to-global character¹, and are often a matter of brute force computation, rather than board understanding.

At the same time, partly due to its local-to-global nature *Go* is almost simultaneous. We make this precise by proving that *Go* has a simple (in a certain mathematical sense) symmetric simultaneization, see Theorem 8. The construction of a particular such simultaneization, called *SGo*, is the main thrust of this note. *SGo* is a combinatorial symmetric simultaneous game, which is playable on a standard *Go* board, although we need a third type of stones that we call red. It is in some ways a simpler game than *Go* as it loses the Komi, Ko and suicide rules, and it has just two rules stone placement and capture (aside from end of game rule). Its game dynamics are very similar to the classical *Go*.

The idea for *SGo* is loosely inspired by quantum mechanics. The rough idea is to branch the game state whenever the moves of the two players do not commute or otherwise interfere in classical *Go*. This is part of a more general theory of universal simultaneization of certain sequential games developed in Section 4.3, and this may be of independent interest. However, the universal simultaneization generally has uninteresting dynamics, cf. Section 2.1. To get interesting dynamics in the case of *SGo*, we also have “objective reduction of the quantum state”- it is

¹In the sense that if a player engages in a Ko fight, they must make moves that are strategically disconnected.

a deterministic, partial collapse of branches based on objective board information. The term objective reduction is coined by Penrose, see for instance [4]. Of course, the word quantum is used figuratively, there is no unitary time evolution, complex amplitudes, and no probabilistic outcomes. The game is deterministic and it would not be appropriate to call it “quantum Go”.

Using the famous idea of Nash in [3], we show that *SGo* has a mixed equilibrium strategy to draw on average. That said finding this even approximately is likely only practical on very small boards, due to the unreal size of the strategy space. We can also ask if *SGo* has a strategy to draw not just draw on average, this is true for instance on a 2x2 board.

The classical game *Go* has inspired a wealth of research on combinatorial game theory, see for instance [1], [5]. It also lead to Conway’s theory of surreal numbers. There is also recent research, see for instance [2] on simultaneous combinatorial games. We expect that *SGo* will provide a simple but rich example that can help with further study of such games.

2. GAME RULES

The game *SGo* is initialized with an empty standard *Go* board, with horizontal/vertical lines whose intersections we just call intersections. There are two players that we call Black and White. As in *Go*, players Black and White move on a given turn by placing a black respectively white stone on an empty intersection on the board, but in *SGo* they do this on the same turn without knowledge of the other’s move on this turn. We allow pass moves as in *Go*. Simultaneous action requires modifying the placement and capture rules of *Go*, while removing the Komi rule, Ko, and suicide rules. We also need to introduce a third type of stones, called red, which may be thought of as simultaneously white and black.

2.1. Motivation for the rules. Before we proceed to the main rules, we briefly overview the motivation, explaining that the rules are not totally ad hoc. The construction of *SGo* is motivated by the following:

- (1) It forms a simple simultaneization of *Go*, in the precise sense of Theorem 8. Being a simultaneization loosely means that *SGo* “embeds” into the universal simultaneization of *Go*. For example, the “causal reduction rule” in Section 2.5 is forced by the necessity of satisfying this condition.
- (2) It is playable on a standard *Go* board, which in particular requires us to represent simultaneously moves to the same position, and we choose to do this by introducing red stones. In this case there is no real choice, any two faithful ways to represent such simultaneous action are clearly equivalent. We are representing the branched state where White moves to an intersection and Black passes, and Black moves to that intersection and White passes.
- (3) We want the minimum amount of rules, in particular we don’t want ad hoc rules dealing with Ko type situation.
- (4) The last point is at the moment less precise, but would be interesting to study more closely in the future. We want *SGo* to have similar game dynamics to the standard *Go*, that is with nearly optimal play the positions should tend to become very sharp, that is have high sensitivity. This is far from the case for the universal simultaneization $\text{Sim}(Go)$ of *Go*, as

considered in Section 4.3. For $\text{Sim}(Go)$ games tend to be drawn due to players lacking forcing options.

To achieve these dynamics, we want to “optimally” prune branched histories resulting from simultaneous action. This has the effect of giving players very forcing options, see Section 3.2 for some examples. We do this via our capture/placement resolution rules, which use objective board information. But this is not without constraints, we need to prune branches so that the game stays symmetric, while having a well defined deterministic state evolution, as elaborated by Lemma 1. We also cannot prune too much, for example in the suicidal capture condition in Definition 2 we intentionally preserve certain branching to avoid the Ko type infinite loops, also see Remark 2.

2.2. Some definitions. Given a board state S , that is any (at the moment no legality restrictions) collection of white, black, red stones on the board, let $G(S)$ denote the graph with vertices stones, and an edge between vertices whenever the corresponding stones are horizontally or vertically adjacent. (Meaning they are adjacent and on the same horizontal/vertical line.) From now on adjacency always means this type of adjacency. In what follows the terms stones and vertices may be used synonymously.

A **unicolor component** C is a connected component of the full (a.k.a. induced) subgraph of $G(S)$ whose vertices are either all the black, or all the white stones. (Full subgraph means that all the edges of the original graph, between any pair of vertices of the subgraph, are included.) In this case we call C **black**, respectively **white**. We may less often refer to **red components**, analogously defined as connected components of red stones.

As in classical Go , a connected component C **has no liberties** if it is not adjacent to an empty intersection (this is short for C does not contain a stone adjacent to an empty intersection). We will also call C a **trapped component**.

We define stones **adjacent** to C as those stones whose corresponding vertices are connected by an edge to C in $G(S)$.

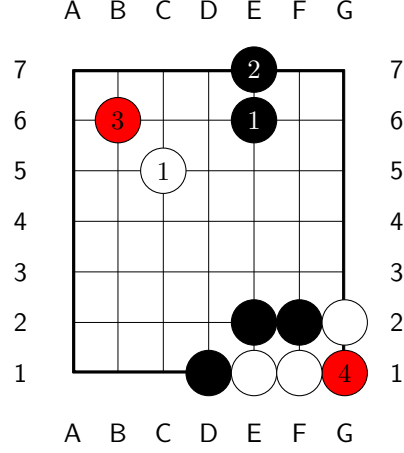
A white/black stone on the board is called a **trapping stone** if it is adjacent to a trapped unicolor component of opposite color.

2.3. Stone placement resolution rule of SGo . If the players move to the same intersection i on the board, we place a stone there whose color is resolved in the following decreasing order of priority:

- (1) If either a white or black stone placed at i is trapping, then a red stone is placed at i . We call this a **trapping move**.
- (2) If the intersection i is adjacent to a black stone, and not a white stone, a black stone is placed. If the intersection is adjacent to a white stone and not a black stone, a white stone is placed.
- (3) Otherwise, a red stone is placed.

Remark 1. We could do something simpler and always place a red stone in the case of simultaneous movement to i . However, as discussed in Section 2.1 we want to aggressively prune some branched histories, using objective board information.

To the right is an example of placement resolution. On move two both players move to E7, the resulting stone resolves to black by our rules. Note that the stone placed on move 4 does not resolve to white, as it is a trapping move.



2.4. Capture rule of SGo. After both White and Black move on the current turn numbered n , using the stone placement resolution rule above we get some board state S .

Definition 1. We say that a black/white/red component C is **trapped on turn n** , if C is trapped and some stone adjacent to C or in C was placed on turn n . A stone is called **trapped** on turn n , if it is an element of C as above. We say that a red stone is **trapped by white respectively black on turn n** , if replacing it by a black respectively white stone it is trapped on turn n .

Definition 2. A white component C is resolved as **captured on this turn** if the following is satisfied:

- (1) C is trapped on this turn.
- (2) If C is adjacent to a red stone that is itself adjacent to another trapped component, then that component is white. (Without this condition Lemma 1 does not work, and the game evolution will be ill defined.)
- (3) One of the following holds:
 - (a) No black or red stone adjacent to C is trapped by white on this turn.
 - (b) (Suicidal capture) No stone in C is created on this turn. If a stone in C is adjacent to a stone trapped on this turn, then that stone is created on this turn and is not red.

Remark 2. The suicidal capture property may appear complicated, but is driven by our requirements, as outlined in Section 2.1. For example, if we remove the part of the condition “is not red”, then Ko-style infinite loops will become possible later. If we remove the condition “no stone in C is created on this turn” the game evolution will be ill defined, specifically Lemma 1 will not work.

In the case C is captured.

- The stones of C are removed as prisoners of Black as in Go.
- Any red stones adjacent to C resolve as black. (Concretely, they are replaced by black stones.)

If C is a black component the process is naturally mirrored. The above capture rule is invoked recursively until no captured on this turn n , unicolor components remain.

2.5. Causal reduction. At the end of the capture procedure, we further simplify the position as follows. If any trapped red component C is adjacent only to white stones, and none of these white stones are trapped then all stones of C are replaced by white stones. This is naturally mirrored in the case C is adjacent only to black stones. The motivation for the term ‘causal reduction’ is explained in Section 3.2.1.

Remark 3. *This additional rule is logically necessary as we want SGo to be a simultaneization of Go, see Section 4.3. Without this rule, condition 3 of Definition 8 will not be satisfied, as the image of natural map ρ in the proof of Theorem 8 would contain a position that is not legally reachable.*

2.6. Stage one reduction. The above capture procedure followed by causal reduction is called **stage one reduction** and the resulting board state will be denoted by $R_n(S)$.

Lemma 1. *The stage one reduction is well defined, that is independent of the order in which components are captured.*

Proof. Suppose we have two unicolor components C_0, C_1 captured on the given turn. By assumption if a red stone is adjacent to both C_0 and C_1 then C_0 and C_1 have the same color. But in that case it clearly does not matter in which order they are captured.

Thus suppose that no red stone is adjacent to both C_0 and C_1 . If C_0 and C_1 have the same color they cannot be adjacent, that is there is no pair of stones, one in C_0 the other in C_1 that are adjacent. We now show that if they have opposite color then they are also not adjacent. Given this, if we apply capture rule to the group C_0 and then to the group C_1 , this results in the same board state as first applying the capture rule to the group C_1 and then the group C_0 .

We have the following possibilities. If C_0, C_1 both satisfy condition 3a, then they are trivially not adjacent. The same happens if C_0 satisfies the former condition and C_1 satisfies condition 3b. Suppose lastly that C_0, C_1 both satisfy condition 3b. And suppose by contradiction that they are adjacent. Then C_1 is adjacent to a stone in C_0 , which by assumption is trapped on this turn. Condition 3b applied to C_1 implies that the latter stone is created on this turn. But by assumptions no stone of C_0 is created on this turn. So we have a contradiction, and so C_0, C_1 are not adjacent.

To prove the general case, we proceed by induction. Let $S(n)$ be the sentence: for a collection $\{C_0, \dots, C_n\}$ of captured unicolor components, the result of stage one reduction does not depend on the order of C_i . $S(1)$ was shown to hold. We show $S(n) \implies S(n+1)$. Let $\{C_0, \dots, C_{n+1}\}$ be a given ordered collection of captured unicolor components, and let ρ be a permutation of the set $\{0, \dots, n+1\}$. By the argument above and using a sequence of adjacent transpositions, the stage 1 reduction applied to $\{C_{\rho(0)}, \dots, C_{\rho(n+1)}\}$ (respecting the given order) is the stage 1 reduction applied to $\{C_0, C_{\rho'(1)}, \dots, C_{\rho'(n+1)}\}$ for some permutation ρ' fixing 0. Using the induction hypothesis on $\{C_{\rho'(1)}, \dots, C_{\rho'(n+1)}\}$ we get that the latter stage 1 reduction is the stage 1 reduction applied to the ordered collection $\{C_0, \dots, C_{n+1}\}$. $\square$

If there are no unicolor components trapped on the turn $n - 1$ or lower, the turn is resolved. Otherwise, we recursively define higher stage reduction as follows.

Define stage k reduction by the condition that the board state after stage k reduction, denoted by $R^k(S)$, satisfies:

$$\begin{aligned} R^1(S) &= R_n(S). \\ R^k(S) &= R_{n-k+1}(R^{k-1}(S)). \end{aligned}$$

If there are no unicolor components trapped on the turn $n - k$ or lower, the capture stage is resolved after stage k reduction.

Remark 4. *In practice the resolution of most turns is easy to work out mentally, we will give some examples of such “nested entanglement states” further ahead. Of course having a computer work out turn resolution automatically is very helpful.*

2.7. End of the game. An *SGo* game ends if one of the following holds:

- The players agree to end the game, in this case they may also agree to remove some stones on the board as prisoners for either player. (As is common in standard play of *Go*.) Each black or white prisoner gives one point to the player who holds it, while red stone prisoners have no value, this is by consistency with the other game rules, as red stones can’t actually be captured they can only be converted by the causal reduction rule. This is the main way of ending the game.
- The board position repeats two times in a row. For example, both players perform a single stone suicide on the same turn twice in a row, or pass twice in a row. (This is meant to ensure that a winning player can actually force a win when the opponent is stalling.)

We then decide the winner using Chinese *Go*-style rules (this is for simplicity). So the winner is decided by the integer sum of the prisoner count and area count. We now explain the latter.

We say that an empty intersection i on the board **reaches black respectively white** if there is a connected component of empty intersections (as usual meaning a connected component of the graph with vertices empty intersections on the board and edges adjacencies) that contains i and contains an intersection adjacent to a black stone respectively white stone.

We say i is **surrounded by black** if:

- i reaches black.
- i does not reach white. (To emphasize it could reach a red stone.)

We define surrounded by white analogously. Then in the case of player Black the area count is the sum of:

- (1) The number of black stones on the board.
- (2) The number of red stones adjacent to an empty intersection surrounded by black.
- (3) The number of empty intersections surrounded by black.

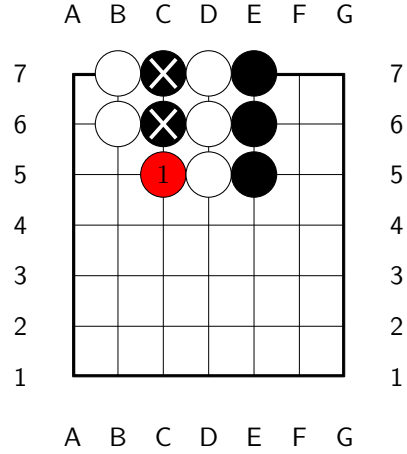
See Section 3.4 for an example.

A draw is now possible, although compared to chess it must be much less likely, (on a typical 13×13 or 19×19 go board).

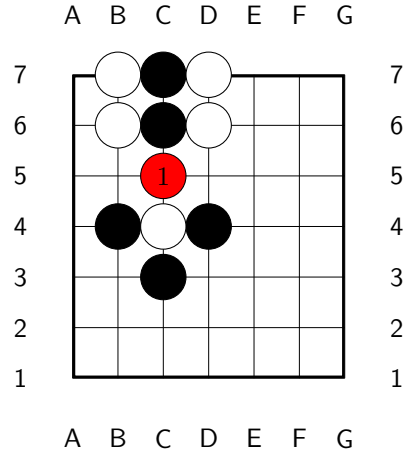
3. EXAMPLES

3.1. Capture and non-capture moves. Here are some examples and non-examples of capture.

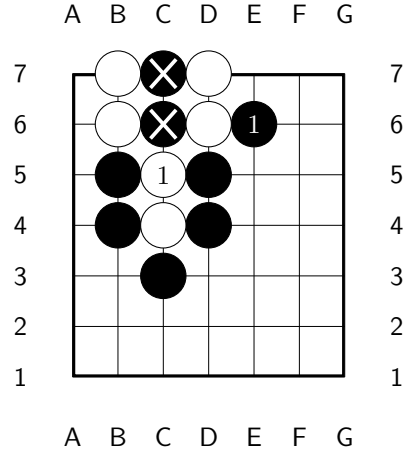
On move 1 both Black and White move to C5. As this is a capture move, C5 will resolve to white at the end of the turn.



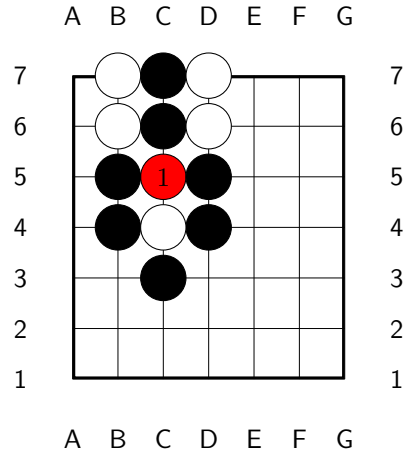
Not a capture move, as the created red stone is adjacent to an opposite-colored trapped component. This should make intuitive sense since any capture would be ambiguous from a classical *Go* perspective.



White's move is a capture move, as Condition 3b is satisfied. This is consistent with classical *Go* behavior that suicidal captures resolve to capture. Formally, this is forced by the condition that *SGo* is a simultaneization of *Go*, Theorem 8.

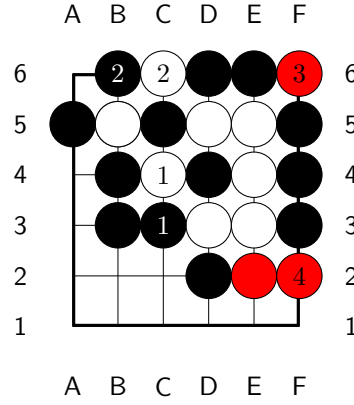


Not a capture move, now Condition 3b is not satisfied, since the stone placed on this turn is red.

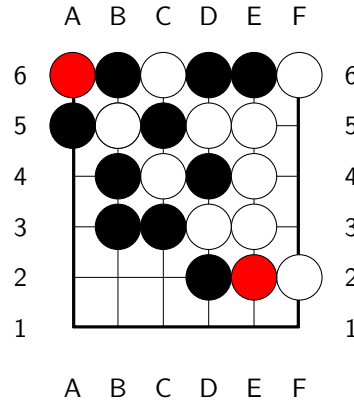


3.2. More complex examples. In the first example White needs only play with a pure strategy (and still be ostensibly winning). In sharper, more complex positions, the players may need mixed strategies.

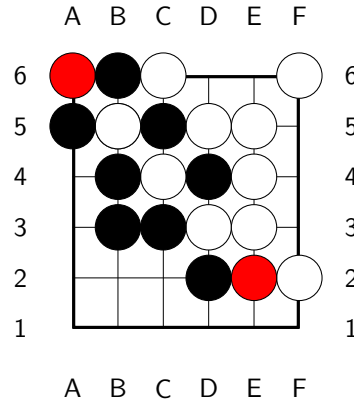
In the position on the right, on turn 1, White attempts capture at C4, while Black attempts capture at C3. Nothing is captured on turn 1 according to our rules, and after turn 2 the position becomes an example of a “nested entangled state”. A possible continuation is indicated. Note that F6 is not a capture for White under our rules, similar to the last example of the previous section. But F2 is a capture move for White since condition **3a** is satisfied.



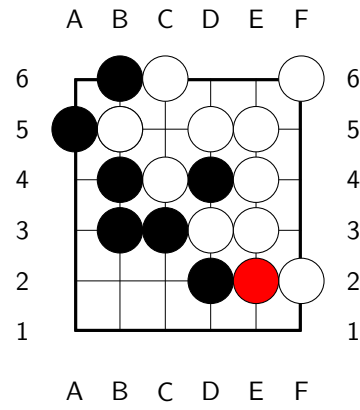
At the end of turn 4, stage one resolution is depicted on the right.



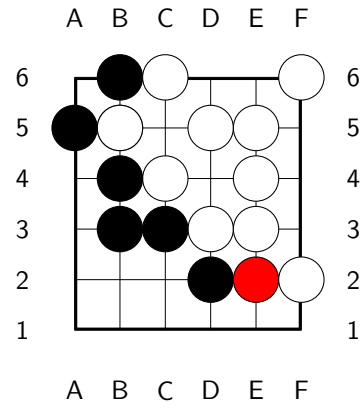
Stage two resolution.



Stage three resolution.

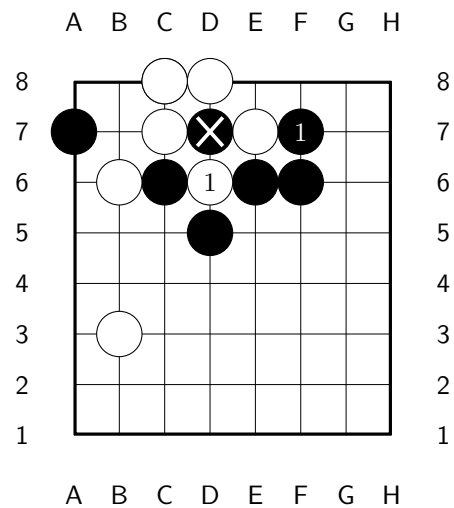


Stage four and final resolution of turn 4.

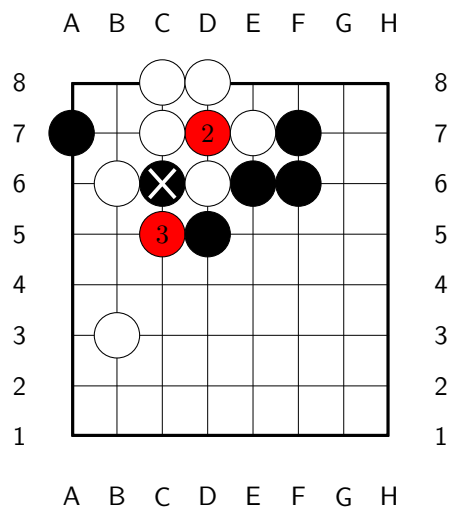


3.2.1. *Ko situation.* Below is a *Ko* type situation. Recall that we don't have a *Ko*-rule, but it is now unnecessary, since the board state cannot immediately loop by re-

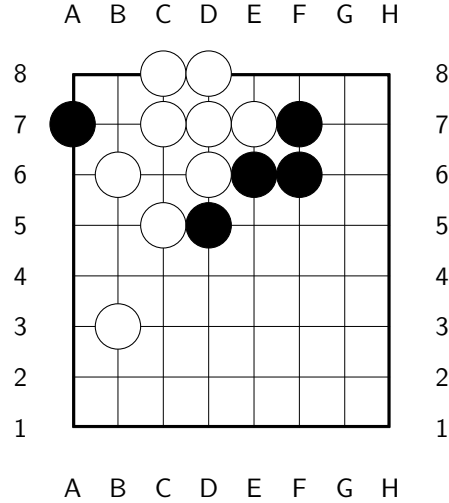
capture. White's D6 is a capture move.



Here is a possible continuation.

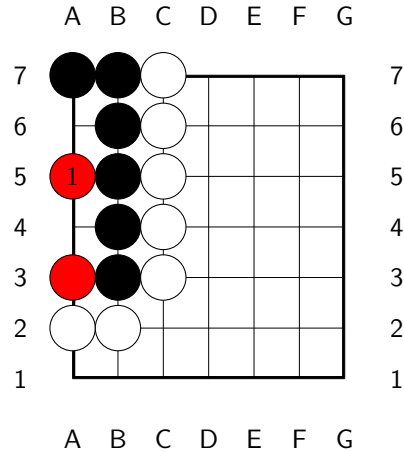


At the end of turn 3 we get the following position. The red stone at D7 is converted to white by the causal reduction rule. This should make sense from the perspective of viewing SGo as Go with quantum-like features. A red stone at D7 is meant to represent both black and white, but if it were black, the resulting position would be illegal in Go, so it must actually be white. (This is the meaning behind the term causal reduction.)

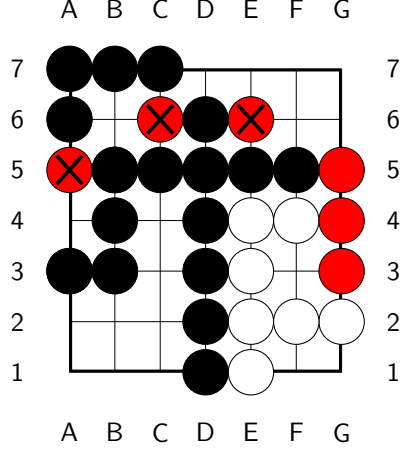


3.3. Life and death. Life and death is positionally very similar to *Go*, here is an example.

The black group is alive, that is impossible to kill.



3.4. End game position. In the diagram below, the players have agreed to end the game. The red stones at *e6*, *c6*, *a5*, are agreed to be prisoners of Black, (the causal reduction rule allows black to consume them if the game was continued, but they are not given a point value). Assuming there are no other prisoners, White has $7 + 3$ points and Black has $7 \cdot 7 - (7 + 3) - 3$ points.



4. SGo AS A SYMMETRIC SIMULTANEOUS COMBINATORIAL GAME

We will use the formalism of automata as it is very flexible. The goal is to introduce symmetric simultaneous combinatorial games.

Definition 3. A combinatorial simultaneous game (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \sqcup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the **pseudo pass move** (it has multi grading both 0 and 1).
- Games states decompose as

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where $B(G)$ is some set (for example it might be the set of board states).

- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in B(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) If $s \in B(G) \subset S(G)$, then $E(s, m) = (s, m)$ for all $m \in M(G)$ s.t. $E(s, m)$ is defined.
- (3) If s is not final $E(s, 0)$ is defined.

- (4) If $(s, m) \in B(G) \times M(G)$, then $E((s, m), m') \in B(G)$ if defined, and is undefined if the grading of m is the grading of m' (and if neither m nor m' are 0).
- (5) Suppose that $m_0 \in M_0(G)$, $m_1 \in M_1(G)$, $s \in B(G)$, $E(s, m_0)$ is defined and $E(s, m_1)$ is defined then $E(s, m_0, m_1)$ is defined.
- (6) For s, m_0, m_1 as in the previous property if $E(s, m_0, m_1)$ is defined then so is $E(s, m_1, m_0)$ and

$$E(s, m_0, m_1) = E(s, m_1, m_0).$$

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

We say that G is **finite** if $S(G)$, $M(G)$ are finite, and all game histories have bounded length, where a game history is a sequence $h = \{m_i\}_{i=1}^{i=n}$ of moves s.t.

$$\forall 1 \leq k \leq n \ E(q_0, m_1, \dots, m_k) \text{ is defined.}$$

Note that any game history of length $2k$ can be rewritten in the form $h = \{m_i^0, m_i^1\}_{i=1}^{i=k}$, with $m_i^j \in M_j(G)$.

It is our convention that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(q_0, m_1, \dots, m_k)$, for some m_i . For future use set $f(h) = E(q_0, m_1, \dots, m_n)$.

The above notion is more interesting when G also satisfies a certain symmetry between the two players.

Definition 4. We say that a combinatorial simultaneous game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{S} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$, and satisfying $\mathcal{S}^2 = \text{id}$, $\mathcal{S}(0) = 0$.
- (2) There is a bijection $R : B(G) \rightarrow B(G)$, satisfying $R^2 = \text{id}$ s.t. for all $s \in B(G)$, all $m \in M(G)$, and for $\mathcal{S}$ as above we have:
 - $R \circ E(R(s), \mathcal{S}(m)) = E(s, m)$, whenever both sides are defined. Furthermore, if one side is defined then so is the other.
 - R also induces a bijection from $W_0(G)$ onto $W_1(G)$ and $D(G)$ onto $D(G)$.

Lemma 2. SGo is an ssc game, that is a symmetric simultaneous combinatorial game.

Proof. Set $G = SGo$, with the latter described in Section 2. The set $M_0(G)$ corresponds to possible moves of Black and $M_1(G)$ the moves of White, as in the game description, with the pseudo pass move 0 the standard pass move.

The state space is:

$$S(G) = B(G) \sqcup B(G) \times M(G),$$

where an element of $B(G)$ is a board state and its game history $h = \{(m_i^0, m_i^1)\}$ with $m_i^j \in M_j(G)$. Here the history is obeying the rules of Section 2.

The evolution map E is then naturally defined. For example if $s \in B(G)$ and $m_0 \in M_0(G)$, $m_1 \in M_1(G)$ are legal moves (obeying the rules of Section 2) then the board state corresponding to $E((s, m_0), m_1) \in B(G)$ is defined to be the resolution of the board position after applying the rules of Section 2, once Black's move m_0 and White's move m_1 are reflected on the board.

To see that G is symmetric, let $R : B(G) \rightarrow B(G)$ be the map that interchanges the black and white colors of the stones of the board states, leaving red stones invariant. And let $\mathcal{S} : M(G) \rightarrow M(G)$ interchange black and white moves, i.e. we have a natural isomorphism $M_0(G) \rightarrow M_1(G)$, which then induces $\mathcal{S}$. Then clearly the conditions of Definition 4 will be satisfied. $\square$

4.1. Solvability. A theorem of Zermelo [6] says that for every finite sequential game either Black or White has a winning strategy or there is a draw strategy. Clearly SGo cannot have a winning strategy, furthermore unless we put an upper bound on the length of a game, or on the number of position repetitions an SGo game length can be infinite.² So we will assume there is such a bound, which will make SGo a finite game.

Question 1. *Does SGo have a strategy to draw for any board size?*

Note that the answer is yes on a 2x2 board as is easy to see. Of course we have combinatorial simultaneous games with no strategy to draw, the classical example is rock-paper-scissors. However, it does have a strategy to draw on average, which is a trivial instance of the existence of a Von Neumann–Nash equilibrium, for “matrix payoff games”. SGo is not a matrix payoff game, but Nash’s proof of the equilibrium theorem readily extends to this case. In general:

Theorem 3. *Assigning any payoff for a win and draw, a finite simultaneous game G in the sense above has a Nash equilibrium.*

Proof. For $s \in B(G)$ set

$$M_i(s) = \{m \in M_i(G) \mid E(s, m) \text{ is defined}\}.$$

The space of strategies of player P_i is then:

$$\mathcal{P}_i = \prod_{s \in B(G)} \Delta(M_i(s)),$$

where $\Delta(M_i(s))$ is the space of probability distributions over the finite set $M_i(s)$. Since $B(G)$ is also finite, the joint strategy space $\mathcal{P}_0 \times \mathcal{P}_1$ is a compact convex subspace of $\mathbb{R}^N$, for some very large N .

The expected payoff function on $\mathcal{P}_0 \times \mathcal{P}_1$ is continuous, due to finiteness of G .³ Using this, as in the argument of Nash, we get that the multi valued function ϕ , assigning to a joint strategy its set of countering strategies, has a closed graph. Furthermore, the image by ϕ of any point is non-empty and convex.

Then as argued by Nash [3] using Kakutani’s fixed point theorem we get existence of an equilibrium point. $\square$

Corollary 4. *A finite ssc game G has a strategy to draw on average, and in particular SGo has such a strategy.*

Proof. Assign 1 for a win, -1 for a loss and 0 to a draw. Then it is enough to show that for any Nash equilibrium strategy the expected payoff is 0 for both players.

Suppose otherwise, and suppose without loss of generality that

$$(\mathcal{D}_0, \mathcal{D}_1) \in \mathcal{P}_0 \times \mathcal{P}_1$$

is an equilibrium joint strategy where Black has a positive expected payoff.

²This is also true for Go and Chess.

³Possibly finiteness condition can be relaxed.

Let $R, \mathcal{S}$ be the symmetry operators as in the proof of Lemma 2. Naturally applying these operators to the joint strategy $(\mathcal{D}_0, \mathcal{D}_1)$ we get that there is an equilibrium joint strategy $(\mathcal{D}'_0, \mathcal{D}'_1)$ where White has positive expected payoff. But this is absurd. $\square$

4.2. Symmetric combinatorial sequential games. Combinatorial sequential games have a beautiful Conway recursive definition. However, for our purpose it is convenient to interpret them in terms of finite automata.

Definition 5. A combinatorial sequential game (with a pseudo pass move) G with 2 players P_0, P_1 consists of:

- A deterministic automaton G , whose set of states is a set $S(G)$ called the set of game states of G .
- The alphabet (that is the set of moves) is a $\mathbb{Z}_2$ graded finite set $M(G) = M_0(G) \cup M_1(G)$: interpreted as possible moves of the players.
- $M_0(G) \cap M_1(G)$ has one move denoted as 0 and called the **pseudo pass move** (it has multi grading both 0 and 1).
- A $\mathbb{Z}_2$ graded set of games states $S(G) = \text{States}(G) = S_0(G) \sqcup S_1(G)$, we let $\deg : S(G) \rightarrow \mathbb{Z}_2$ denote the associated function.
- Nonempty subset $F(G) = W_0(G) \sqcup W_1(G) \sqcup D(G) \subset S(G)$ of final (accepting) states, which are understood as states where P_0 wins, P_1 wins or both players draw, respectively.
- A distinguished initial state $q_0 \in S_0(G)$, understood as the state in which the game is initialized.
- A partial game evolution function:

$$E = E_G : S(G) \times M(G) \rightarrow S(G),$$

which has the properties:

- (1) $E(s, m)$ is undefined for s in one of the final states.
- (2) $E(s, m)$ is defined only if m has the same grading as s or if $m = 0$.
- (3) If $E(s, m)$ is defined then $\deg(E(s, m)) = \deg(s) + 1$.
- (4) If s is not a final state then $E(s, 0)$ is defined.
- (5) If $E(s, 0) = E(s', 0)$ then $s = s'$.

We recursively define:

$$E(s, m_1, \dots, m_k) = E(E(s, m_1, \dots, m_{k-1}), m_k).$$

As before it will be understood that $S(G)$ is the set of legally obtainable positions, that is positions of the form $E(s, m_1, \dots, m_k)$, for some m_i .

Example 5. Note that Go has a pseudo pass move, but chess does not, however chess can be augmented with a pseudo pass move, by also adjusting the rules so that king capture is legal and ends the game, while also removing the check rules. This does not change the early-mid game dynamics at all, but the end game does significantly change, with more tie possibilities (mainly, if a player is in Zugzwang they may just pass). We will call this **augmented chess** denoted by $\widetilde{\text{chess}}$.

Definition 6. We say that a combinatorial sequential game G is **symmetric** if the following conditions are satisfied.

- (1) There is a bijection $\mathcal{S} : M(G) \rightarrow M(G)$ taking $M_0(G)$ onto $M_1(G)$ and satisfying $\mathcal{S}^2 = \text{id}$, $\mathcal{S}(0) = 0$.

- (2) There is a bijection $R : S(G) \rightarrow S(G)$, satisfying $R^2 = \text{id}$ and taking $S_0(G)$ to $S_1(G)$ s.t. for all $s \in S(G)$, all $m \in M(G)$ and for $\mathcal{S}$ as above we have:

$$R \circ E(R(s), \mathcal{S}(m)) = E(s, m),$$

whenever both sides are defined. Furthermore, if one side is defined then so is the other.

Example 6. Of course $\widetilde{\text{chess}}$ and classical Go are examples of symmetric games. In case of Go, $\mathcal{S}$ interchanges black and white moves and R likewise interchanges the color of the stones on the board and shifts the initiative (it is easy to see that this does result in a legally reachable position).

It will be the assumption from now on that all sequential games that appear are symmetric (with $\mathcal{S}$, R implicitly fixed).

4.3. Universal simultaneization. Let G be sequential game (by our standing assumption with a pass move 0 and symmetric). Let $\{m_i\}$ be a sequence of game moves. If some $m_i = 0$ and $m_k \neq 0$ for some $k > i$, then we say that it is an **interior pass move**. We say that a finite sequence $\{m_i\}$ of moves of G is **pseudo legal** if it becomes a game history after adding or removing some interior pass moves from the sequence. It is not hard to see that given a pseudo legal sequence h , there is a unique minimal length game history as above, and we denote it by $\text{leg}(h)$.

For $a \in S(G)$ if there exists an $a' \in S(G)$ s.t. $E_G(a', 0) = a$ we set $\bar{a} = a'$, otherwise set $\bar{a} = E_G(a, 0)$. Property 5 in Definition 5 insures that it doesn't matter which a' is chosen when it exists. For $a \in S(G)$ and $m \in M(G)$ set

$$a \# m = \begin{cases} E_G(a, m), & \text{if this is defined} \\ E_G(\bar{a}, m), & \text{if this is defined} \\ \text{undefined}, & \text{otherwise.} \end{cases}$$

Define $P(a, m_0, m_1) \in 2^{S(G)}$ by the following prescription in decreasing priority, where the expression is defined if only if all the summands are defined.

- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1 \# m_0$ (note that if $m_i = 0$ then the pair of elements coincide so that $P(a, 0, 0) = a \# 0 \# 0$.)
- $P(a, m_0, m_1) := a \# m_0 \# m_1 \cup a \# m_1$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1 \# m_0$.
- $P(a, m_0, m_1) := a \# m_0 \cup a \# m_1$.
- $P(a, m_0, m_1)$ is undefined.

We then get a partial function

$$P : S(G) \times M_0(G) \times M_1(G) \rightarrow 2^{S(G)}.$$

Definition 7. Let G be a sequential game. Its **universal simultaneization** is the simultaneous game $\text{Sim}(G) = G'$ satisfying the following.

- (1) $B(G') = 2^{S_0(G)}$. (The sub elements of $s \in \mathfrak{s} \in B(G')$ will be called **state branches**.)
- (2) The initial state $q'_0 = \{q_0\}$.
- (3) $M(G') = M(G)$ as $\mathbb{Z}_2$ graded sets.
- (4) Let $\mathfrak{s} \in B(G')$, and $m_i \in M_i(G')$ then $E_{G'}(\mathfrak{s}, m_i)$ is defined iff for each $s \in \mathfrak{s}$ s is not final and $s \# m_i$ is defined.

(5) For $\mathfrak{s}$ and m_i as just above:

$$E_{G'}(\mathfrak{s}, m_0, m_1) := \cup_{s \in \mathfrak{s}} \{P(s, m_0, m_1)\},$$

(undefined if some $P(s, m_0, m_1)$ is undefined).

(6)

$$\mathfrak{s} \in F(G') \iff \exists s \in \mathfrak{s} \ s \in F(G).$$

Note that $\text{Sim}(G)$ is uniquely defined by the above conditions up to specifying the sets $W_1(\text{Sim}(G))$, $D(\text{Sim}(G))$; these will not play a role but we can specify them as follows:

•

$$\mathfrak{s} \in W_0(G') \iff \exists s \in \mathfrak{s} \ (s \text{ is winning for } P_0) \wedge \nexists s \in \mathfrak{s} \ (s \text{ is winning for } P_1).$$

• Similarly,

$$\mathfrak{s} \in W_1(G') \iff \exists s \in \mathfrak{s} \ (s \text{ is winning for } P_1) \wedge \nexists s \in \mathfrak{s} \ (s \text{ is winning for } P_0).$$

• In all other cases, $\mathfrak{s} \in D(G')$.

Lemma 7. $\text{Sim}(G)$ is symmetric.

Proof. Set $G' = \text{Sim}(G)$, and let $\mathcal{S}, R$ be the symmetry operators of G . Set $\tilde{\mathcal{S}} : M(G') \rightarrow M(G')$ to just be $\mathcal{S}$. For

$$\{s_0, \dots, s_k\} = \mathfrak{s} \in B(G'),$$

set $\tilde{R}(\mathfrak{s}) = \{R(s_0), \dots, R(s_k)\}$. For

$$(\mathfrak{s}, m) \in B(G') \times M(G') \subset S(G'),$$

set $\tilde{R}(\mathfrak{s}, m) = (\tilde{R}(\mathfrak{s}), \tilde{\mathcal{S}}(m))$.

The needed symmetry properties for the maps $\tilde{\mathcal{S}}, \tilde{R}$ are then verified using the symmetry properties of $\mathcal{S}, R$ as follows. It is easily seen that $\overline{R(a)} = R(\bar{a})$, hence

$$R(a) \# S(m) = \begin{cases} E_G(R(a), S(m)) = R \circ E_G(a, m), & \text{if defined} \\ E_G(R(\bar{a}), S(m)) = R \circ E_G(\bar{a}, m), & \text{if defined} \end{cases}$$

and so

$$(1) \quad R(a) \# S(m) = R(a \# m),$$

furthermore if one side is defined then so is the other. Suppose that $\mathfrak{s} \in B(G')$ and $m_i \in M_i(G')$ then we get:

$$\begin{aligned} \tilde{R}(E_{G'}(\tilde{R}(\mathfrak{s}), \tilde{\mathcal{S}}(m_0), \tilde{\mathcal{S}}(m_1))) &= \tilde{R} \circ \cup_{s \in \mathfrak{s}} \{P(R(s), S(m_0), S(m_1))\} \\ &= \tilde{R} \circ \cup_{s \in \mathfrak{s}} \{R \circ P(s, m_0, m_1)\} \text{ using (1)} \\ &= \cup_{s \in \mathfrak{s}} \{P(s, m_0, m_1)\} \\ &= E_{G'}(\mathfrak{s}, m_0, m_1), \end{aligned}$$

furthermore if one side is defined then so is the other. This verifies part of the symmetry property, specifically property 2 of Definition 6 is verified for $\mathfrak{s} \in B(G') \times M(G')$, the rest is similar. $\square$

Definition 8. Let G_0 be a sequential game. We say that G_1 is a **simultaneization** of G_0 if the following is satisfied.

(1) G_1 is a symmetric simultaneous game.

- (2) $M(G_1) = M(G_0)$ as $\mathbb{Z}_2$ graded sets.
- (3) There is a map $\rho : B(G_1) \rightarrow B(\text{Sim}(G_0))$ satisfying $\rho(q_0) = q'_0$ and s.t. for $s \in B(G_1)$, $E_{G_1}(s, m_0, m_1)$ is defined iff $E_{\text{Sim}(G)}(\rho(s), m_0, m_1)$ is defined, and in this case

$$\rho(E_{G_1}(s, m_0, m_1)) \subset E_{\text{Sim}(G)}(\rho(s), m_0, m_1).$$

(This expresses the idea that evolution in G_1 is the evolution in the universal simultaneization with some branch reduction mechanism.)

- (4) $\rho(F(G_1)) \subset F(\text{Sim}(G_0))$.
- (5) We say that a state $s \in B(G_1)$ is **pure** (relative to G_0) if there is finite game history $h_s = \{m_i^0, m_i^1\}$ of G_1 s.t. $f(h) = s$ and s.t. both $h_s = \{m_i^0, m_i^1\}$ and $h'_s = \{m_i^1, m_i^0\}$ are pseudo legal move sequences of G_0 and $f(\text{leg}(h_s)) = f(\text{leg}(h'_s))$. We then ask that: for a pure s $\rho(s)$ is a singleton set consisting of $f(\text{leg}(h_s))$.
- (6) If s is pure and $\rho(E_{G_1}(s, m_0, m_1))$ is a singleton then $\rho(s) \# m_0 \# m_1 = \rho(s) \# m_1 \# m_0$.

Definition 9. Let G_0 be sequential game. We will say that G_1 is a **simple simultaneization** of G_0 if it is a simultaneization and there is a map $\phi : B(G_1) \rightarrow B(G_0)$ s.t.:

- $\phi(q'_0) = q_0$ (for q_0, q'_0 the respective initial states.)
- For $s \in B(G_1)$, $m \in M(G)$: $E_{G_1}(s, m)$ is defined if and only if $\phi(s) \# m$ is defined and s is not final.

Remark 5. There may certainly be other notions of “simplicity”. This particular definition is very tentative.

Let Go be the classical game with Chinese rules and standard pass move, such that the game is ended analogously to the way SGo is ended.

Theorem 8. SGo is a simple simultaneization of Go .

Proof. The first two properties are by construction and Lemma 2. Construct the map ρ as follows. Suppose we are given an SGo history $h = \{m_i^0, m_i^1\}_{i=1}^k$, where m_i^0 is the move of P_0 on turn i , and m_i^1 is the move of P_1 on turn i . We then get $P(\{m_i^0, m_i^1\}) \subset \text{States}(Go)$ defined by recursion

$$P(h) = \bigsqcup_{a \in P(\{m_i^0, m_i^1\}_{i=1}^{k-1})} P(a, m_k^0, m_k^1).$$

$\rho(a) \subset P(\{m_i^0, m_i^1\})$ is then obtained by eliminating certain history branches by way of the SGo turn resolution rules, in the obvious sense. It is then mostly immediate that the remaining properties hold.

To verify the simplicity property, we can do something trivial. Define

$$\phi : B(SGo) \rightarrow \text{States}(Go)$$

by replacing every stone of the corresponding SGo board state s by a black stone, giving us $\phi(s)$. It is then immediate that the simplicity property is satisfied with this choice of ϕ . $\square$

Question 2. Does augmented chess as in Example 5 have a simple simultaneization?

Of course augmented chess has the universal simultaneization $\mathcal{G} = \widetilde{\text{Sim}(\text{chess})}$.

Theorem 9. $\mathcal{G}$ is not a simple simultaneization of $\widetilde{\text{chess}}$.

Proof. Suppose otherwise, and let $\phi : B(\mathcal{G}) \rightarrow S(\widetilde{\text{chess}})$ be such that the simplicity property is satisfied. Let $s_0 \in B(\mathcal{G})$ be the pure state corresponding to the following game sequence where White move is indicated first and using standard (except for the pass move usage) chess algebraic notation $a4, 0; Ra3, 0; Rb3, 0$.

In the position White has the move $m_0 = Rb6$ and Black has the pawn move $m_1 = B5$. These moves interfere and we set $\mathfrak{s}_1 = E_{\mathcal{G}}(s_0, m_0, m_1) = \{a_0, a_1\}$ for distinct a_i .

Then the above queen side White rook is in different locations on the board for a_0 and a_1 , and hence White has no legal rook move in the position $\mathfrak{s}_1$. Hence, White has no legal rook move in position $\phi(\mathfrak{s}_1)$ (it would have to be legal in every branch). On the other hand since we know all the legal moves in the position $\mathfrak{s}_1$, the position of all the pieces of $\phi(\mathfrak{s}_1)$ except the queen side rook and the b pawn are forced by the defining property of ϕ . It is then clear $\phi(\mathfrak{s}_1)$ is not legally obtainable, since there is no possible game history for this rook (it couldn't have been captured for instance since the other pawns in $\phi(\mathfrak{s}_1)$ are in the starting position). And so we have a contradiction. $\square$

The answer to Question 2 should be no for similar reasons, but this is not a simple generalization as there may be some non obvious history reduction mechanism, akin to what is happening in SGo , that yields a simple simultaneization.

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